

ANSWERS

1	(a)	product of (magnitude of one) force and distance between forces reference to <i>either</i> perpendicular distance between forces or line of action of forces & perpendicular distance	M1 A1	[2]
	(b)	(i) 90° (ii) $130 = F \times 0.45$ (allow e.c.f. for angle in (i)) $F = 290\text{ N}$ (allow 1 mark only if angle stated in (i) is not used in (ii))	B1 C1 A1	[1] [2]
2	(a)	(i) e.g. (phase) change from liquid to gas / vapour thermal energy required to maintain constant temperature (do not allow 'convert water to steam') (ii) e.g. evaporation takes place at surface boiling takes place in body of the liquid e.g. evaporation occurs at all temperatures boiling occurs at one temperature	B1 B1 B1 B1	[1] [4]
	(b)	(i) volume = $(\frac{48}{4.5}) = 10.7\text{ cm}^3$ (ii) 1 volume = $10.7 / (6.0 \times 10^{23})$ $= 1.8 \times 10^{-23}\text{ cm}^3$ 2 separation = $\sqrt[3]{(1.8 \times 10^{-23})}$ $= 2.6 \times 10^{-8}\text{ cm}$	A1 A1 A1	[1] [1] [1]
3	(a)	<u>point</u> at which (whole) weight (of body) (allow mass for weight) appears / seems to act ... (for mass need 'appears to be concentrated')	M1 A1	[2]
	(b)	(i) point C shown at centre of rectangle $\pm 5\text{ mm}$ (ii) arrow vertically downwards, from C with arrow starting from the same margin of error as in (b)(i)	B1 B1	[1] [1]
	(c)	(i) reaction / upwards / supporting / normal reaction force friction force(s) at the rod (ii) comes to rest with (line of action of) weight acting through rod allow C vertically below the rod so that <u>weight</u> does not have a moment about the pivot / rod	M1 M1 A1 B1 B1	[3] [2]
4	(a)	point where the weight of an object / gravitational force may be considered to act	M1 A1	[2]
	(b)	product of the force and the <u>perpendicular</u> distance (to the pivot)	B1	[1]
	(c)	(i) 1. sum / net / resultant force is zero 2. net / resultant moment is zero sum of clockwise moments = sum of anticlockwise moments (ii) $W \times 0.2 = 80 \times 0.5 + 70 \times 1.3$ $= 40 + 91$ $W = 655\text{ N}$ (allow 2/3 for one error in distance but 0/3 if two errors) (iii) move pivot to left gives greater clockwise moment / smaller anticlockwise moment or move W to right gives smaller anticlockwise moment	B1 B1 C1 C1 A1 (M1) (A1) (M1) (A1)	[2] [3] [2]

9 (a)	mass is the property of a body resisting changes in motion / quantity of matter in a body / measure of inertia to changes in motion weight is the force due to the gravitational field/force due to gravity or gravitational force Allow 1/2 for 'mass is scalar weight is vector'	B1	B1 [2]
(b) (i)	arrow vertically down through O tension forces in correct direction on rope	B1	B1 [2]
(ii)	1. weight = $mg = 4.9 \times 9.81 (= 48.07)$ $69 \sin \theta = mg$ $\theta = 44.(1)^\circ$ use of cos or tan 1/3 only	C1	C1
	scale drawing allow $\pm 2^\circ$	A1	[3]
	2. $T = 69 \cos \theta$ $= 49.6 / 50 \text{ N}$	C1	A1 [2]
	scale drawing 50 ± 2 (2/2) 50 ± 4 (1/2)		
	correct answers obtained using scale diagram or triangle of forces will score full marks cos in 1. then sin in 2. (2/2)		
10 (a)	the point where (all) the weight (of the body) is considered / seems to act	M1	A1 [2]
(b) (i)	vertical component of $T (= 30 \cos 40^\circ) = 23 \text{ N}$	A1	[1]
(ii)	the <u>sum</u> of the clockwise moments about a <u>point</u> equals the <u>sum</u> of the anticlockwise moments (about the same point)	B1	[1]
(iii)	(moments about A): 23×1.2 (27.58) $= 8.5 \times 0.60 + 1.2 \times W$ working to show $W = 19$ or answer of 18.73 (N)	M1	M1
		A1	[3]
(iv)	$(M = W / g = 18.73 / 9.81 =) 1.9(09) \text{ kg}$	A1	[1]
	(for equilibrium) resultant force (and moment) = 0 upward force does not equal downward force / horizontal component of T not balanced by forces shown	B1	
		B1	[2]
11 (a)	torque of a couple = <u>one</u> of the forces / a force $\times$ distance multiplied by the <u>perpendicular distance between the forces</u>	M1	A1 [2]
(b) (i)	weight at P (vertically) down normal reaction OR contact force at (point of contact with the pin) P (vertically) up	B1	
		B1	[2]
(ii)	torque = 35×0.25 (or 25) $\times 2$ $= 18$ (17.5) N m	C1	A1
			[2]
(iii)	the two 35 N forces are equal and opposite and the weight and the upward / contact / reaction force are equal and opposite	B1	[1]
(iv)	not in equilibrium as the (resultant) torque is not zero	B1	[1]

12 (a)	pressure = force / area (normal to force)	A1	[1]
(b)	molecules/atoms/particles in (constant) random/haphazard motion	B1	
	molecules have a <u>change</u> in momentum when they collide with <u>the walls</u>	M1	
	(force exerted on molecules) therefore force on the walls	A1	
	reference to average force from many molecules/many collisions	A1	[4]
(c)	elastic collision when <u>kinetic</u> energy conserved	B1	
	temperature constant for gas	B1	[2]
13 (a) (i)	resultant force is zero	B1	
	weight of plank + weight of man = $F_A + F_B$		
	or $200 \text{ (N)} + 880 \text{ (N)}$ or $1080 = F_A + F_B$	B1	[2]
(ii)	principle of moments used	C1	
	(anticlockwise moments) $F_B \times 5.0$	C1	
	(clockwise moments) $880 \times 0.5 + 200 \times 2.5$	C1	
	$F_B = (440 + 500)/5.0 = 188 \text{ N}$	A1	[4]
(b)	straight line with positive gradient (allow freehand)	M1	
	start point (0, 100)	A1	
	finish point (5, 980)	A1	[3]
14 (a) (i)	(vertical component = $44 \sin 30^\circ = 22 \text{ N}$)	A1	[1]
(ii)	(horizontal component = $44 \cos 30^\circ = 38(.1) \text{ N}$)	A1	[1]
(b)	$W \times 0.64 = 22 \times 1.60$	C1	
	($W =$) 55 N	A1	[2]
(c)	F has a horizontal component (not balanced by W)		
	or F has 38 N acting horizontally		
	or 38 N acts on wall		
	or vertical component of F does not balance W		
	or F and W do not make a closed triangle of forces	B1	[1]
(d)	line from P in direction towards point on wire vertically above W and direction up	B1	[1]
15 (a)	$p = F/A$	M1	
	use of $m = \rho V$ and use of $V = Ah$ and use of $F = mg$	M1	
	correct substitution to obtain $p = \rho gh$	A1	[3]
(b) (i)	(when h is zero the pressure is not zero due to) <u>pressure</u> from the air/atmosphere	B1	[1]
(ii)	gradient = ρg or $P - 1.0 \times 10^5 = \rho gh$	C1	
	e.g. $\rho g = 1.0 \times 10^5 / 0.75 (= 133333)$		
	$\rho = 133333 / 9.81$		
	$= 14000 (13592) \text{ kg m}^{-3}$	A1	[2]

16 (a) (resultant force = 0) (equilibrium)

therefore: weight – upthrust = force from thin wire (allow tension in wire)

or

$$5.3 \text{ (N)} - \text{upthrust} = 4.8 \text{ (N)}$$

B1 [1]

(b) difference in weight = upthrust or upthrust = 0.5 (N)

$$0.5 = \rho ghA \text{ or } m = 0.5/9.81 \text{ and } V = 5.0 \times 13 \times 10^{-6} \text{ (m}^3\text{)}$$

C1

$$\rho = 0.5/(9.81 \times 5.0 \times 13 \times 10^{-6})$$

C1

$$= 780 \text{ (784) kgm}^{-3}$$

A1 [3]

17 (a) resultant force (in any direction) is zero

B1

resultant moment/torque (about any point) is zero

B1 [2]

(b) (i) force = $33 \sin 52^\circ$ or $33 \cos 38^\circ$
= 26 N

A1 [1]

(ii) 26×0.30 or $W \times 0.20$ or 12×0.40

C1

$$26 \times 0.30 = (W \times 0.20) + (12 \times 0.40)$$

C1

$$W = 15 \text{ N}$$

A1 [3]

(c) (i) $E = \Delta\sigma / \Delta\varepsilon$ or $E = \sigma / \varepsilon$

C1

$$\Delta\sigma = 2.0 \times 10^{11} \times 7.5 \times 10^{-4}$$

$$= 1.5 \times 10^8 \text{ Pa}$$

A1 [2]

(ii) $\Delta\sigma = \Delta F / A$ or $\sigma = F / A$

C1

$$A = 78 / 1.5 \times 10^8 \text{ (= } 5.2 \times 10^{-7} \text{ m}^2\text{)}$$

C1

$$5.2 \times 10^{-7} = \pi d^2 / 4$$

$$d = 8.1 \times 10^{-4} \text{ m}$$

A1 [3]

18 (a)	$\rho = m/V$	C1
	$V = \pi d^2 L/4$ or $\pi r^2 L$	C1
	weight = $2.7 \times 10^3 \times \pi (1.2 \times 10^{-2})^2 \times 5.0 \times 10^{-2} \times 9.81 = 0.60 \text{ N}$	A1
(b)(i)	the point from where (all) the weight (of a body) seems to act	B1
(ii)	$W \times 12$	C1
	$(0.25 \times 8) + (0.6 \times 38)$	C1
	$W = (2 + 22.8) / 12$	A1
	$= 2.1 (2.07) \text{ N}$	
(c)(i)	pressure changes with depth (in water)	B1
	or pressure on bottom (of cylinder) different from pressure on top	
	pressure on bottom of cylinder <u>greater than</u> pressure on top	B1
	or force (up) on bottom of cylinder <u>greater than</u> force (down) on top	
(ii)	anticlockwise moment reduced and reducing the weight of X reduces clockwise moment	
	or anticlockwise moment reduced so clockwise moment	B1
	now greater than (total) anticlockwise moment	

19	resultant force (in any direction) is zero	B1
	resultant torque/moment (about any point) is zero	B1

20 (a)	force $\times$ <u>perpendicular</u> distance (of line of action of force) to/from a point	B1
(b) (i)	$2.4r$ or $(1.2 \times 2r)$ or $(1.2r + 1.2r)$	A1
(ii)	(anticlockwise moment $\Rightarrow$) $6.0 \times r/2 \times \sin \theta$	C1
	$6.0 \times r/2 \times \sin \theta = 2.4r$	A1
	$\theta = 53^\circ$	
(iii)	6.0 N	A1

21 (a)	$\rho = m/V$ or $\rho = m/Ah$	B1
	$p = F/A$ or $p = W/A$	B1
	$p = [\rho Ahg]/A$ or $p = [\rho Vg]/[V/h]$ (so) $p = \rho gh$	A1
(b) (i)	1. weight/gravitational (force) upthrust (force)/buoyancy (force) drag/viscous/frictional (force)/fluid resistance/resistance	B1
	2. weight = upthrust + viscous (force)	B1
(ii)	• decrease in (gravitational) potential energy (of sphere) due to decrease in height (since $E_p = mgh$)	B2
	• increase in thermal energy due to work done against viscous force/drag	
	• loss/change of (total) E_p equal to gain/change in thermal energy	
	Any 2 points.	
(c) (i)	atmospheric pressure = $9.1(0) \times 10^4 \text{ Pa}$	A1
(ii)	$(\Delta)p = \rho g(\Delta)h$	C1
	$(9.15 - 9.10) \times 10^4 = \rho \times 9.81 \times (0.17 - 0.10)$	
	$\rho = 730 (728) \text{ kg m}^{-3}$	A1

22 (a)	ampere kelvin (allow mole, candela) any two correct answers, 1 mark each	B2
(b) (i)	frictional (force)/friction	B1
(ii)	$W \cos 31^\circ \times 3.0$ or 90×6.0 $W \cos 31^\circ \times 3.0 = 90 \times 6.0$ $W = 210 \text{ N}$	C1 A1
(iii)	$X = 90 \sin 31^\circ$ $= 46 \text{ N}$	A1
23 (a)	$P = Fv$ $P = 8.9 \cos 30^\circ \times 0.60$ $= 4.6 \text{ W}$	C1 A1
(b)	$p = F/A$ $F = 8.9 \sin 30^\circ + (0.24 \times 9.81)$ $(= 6.80 \text{ N})$ $A = 6.80 / 3500$ $= 1.9 \times 10^{-3} \text{ m}^2$	C1 C1 A1
(c) (i)	upwards/up	B1
(ii)	the Earth/planet	B1
24 (a) (i)	(vertically) upwards/up	B1
(ii)	increases (with time/velocity/depth)	B1
(b) (i)	for a body in (rotational) equilibrium <u>sum/total</u> of clockwise moments about a point = <u>sum/total</u> of anticlockwise moments about the (same) point	B1 B1
(ii)	$(F_B \times 5.0)$ or (380×2.5) or (750×1.6) $(F_B \times 5.0) = (380 \times 2.5) + (750 \times 1.6)$ $F_B = 430 \text{ N}$	C1 A1
(iii)	taking moments about C: $(380 \times 2.5) = 750 \times (2.0 - x)$ $(2.0 - x) = 1.3$ $x = 0.7 \text{ m}$	C1 A1
or	moments may be taken about other points, e.g. about D: $(380 \times 4.5) + (750 \times x) = 1130 \times 2.0$ $x = 0.7 \text{ m}$	(C1) (A1)
25 (a)	base units: $\text{kg m s}^{-2} \times \text{m}$ $= \text{kg m}^2 \text{ s}^{-2}$	A1
(b) (i)	distance of COG from P (= GP) $= 17 \cos 45^\circ - 4.0$ or $(144.5)^{1/2} - 4.0$ (= 8.0 cm) moment = $0.15 \times 8.0 \times 10^{-2}$ $= 1.2 \times 10^{-2} \text{ Nm}$	C1 A1
(ii)	(line of action of) weight acts through pivot/P or distance between (line of action of) weight and pivot/P is zero (so) weight does not have a moment about pivot/P	B1 B1

26.

(a)	resultant force (in any direction) is zero
	resultant torque/moment (about any point) is zero
(b)(i)	1. $T \sin 53^\circ = 2.4$ $T = 3.0 \text{ N}$
	2. $F = T \cos 53^\circ$ or $F^2 = T^2 - 2.4^2$ $F = 1.8 \text{ N}$
(b)(ii)	$\sigma = T/A$ or $\sigma = F/A$
	$A = \pi d^2 / 4$ or $A = \pi r^2$
	$\sigma = 3.0 \times 4 / [\pi \times (0.50 \times 10^{-3})^2]$ $= 1.5 \times 10^7 \text{ Pa}$
(c)(i)	$h = 75 - 75 \sin 53^\circ = 15 \text{ cm}$
(c)(ii)	$(\Delta)E = mg(\Delta)h$ or $(\Delta)E = W(\Delta)h$
	$(\Delta)E = 2.4 \times 15 \times 10^{-2}$ $= 0.36 \text{ J}$
(c)(iii)	$E = \frac{1}{2}mv^2$
	$0.36 = \frac{1}{2} \times (2.4 / 9.81) \times v^2$
	$v = 1.7 \text{ m s}^{-1}$

27.

(a)	for a body in (rotational) equilibrium
	sum/total of clockwise moments about a point = sum/total of anticlockwise moments about the (same) point
(b)(i)	$(W \times 0.45)$ or (19×1.3) or $(W \times 1.85)$ or (22×2.6)
	$(W \times 0.45) + (19 \times 1.3) + (W \times 1.85) = (22 \times 2.6)$ so $W = 14 \text{ N}$
(b)(ii)	magnitude = $19 + 14 + 14 - 22$ $= 25 \text{ N}$
	direction: vertically upwards
(c)(i)	the extension is zero when the force is zero
	graph is a straight line and (so) Hooke's law obeyed
(c)(ii)	$k = F/x$ or $k = \text{gradient}$
	e.g. $k = 60 / (1.00 - 0.25)$
	$k = 80 \text{ N m}^{-1}$
(c)(iii)	area shaded below graph line between $L = 0.25 \text{ m}$ and $L = 0.75 \text{ m}$

28.

(a)(i)	force $\times$ <u>perpendicular</u> distance (of line of action of force to the point)
(a)(ii)	units: $\text{kg m s}^{-2} \text{ m}$ $= \text{kg m}^2 \text{ s}^{-2}$
(b)	$W = \rho Vg$ or $W = \rho ALg$ $A = 5.2 / (790 \times 2.4 \times 9.81)$ $(= 2.8 \times 10^{-4} \text{ (m}^2\text{)})$ $= 2.8 \times 10^2 \text{ mm}^2$
(c)(i)	(component $=$) $5.2 \sin 56^\circ = 4.3 \text{ (N)}$ or $5.2 \cos 34^\circ = 4.3 \text{ (N)}$
(c)(ii)	$(T \times 2.4)$ or (4.3×1.2) or (4.6×1.8) $(T \times 2.4) + (4.3 \times 1.2) = (4.6 \times 1.8)$ $T = 1.3 \text{ N}$

29.

(a)	(component $=$) $96 \sin 38^\circ = 59 \text{ (N)}$ or $96 \cos 52^\circ = 59 \text{ (N)}$
(b)	(45×2.9) or $(T \times 1.8)$ or (59×1.5) $(45 \times 2.9) = (T \times 1.8) + (59 \times 1.5)$ $T = 23 \text{ N}$

30.

(a)	point where (all) the weight (of the body) is considered/seems to act
(b)(i)	horizontal component of force $= 38 \cos 60^\circ$ or $38 \sin 30^\circ$ $= 19 \text{ N}$
(b)(ii)	$(T \times 1.2)$ or (19×0.9) or 17 $(T \times 1.2) = (19 \times 0.9)$ $T = 14 \text{ N}$
b)(iii)	$F = 45 + 38 \sin 60^\circ$ $= 78 \text{ N}$

31.

(a)	solid straight line drawn between centre of sphere at X and at Y
(b)	$p = mv$ or $0.72 = mv$ $E = \frac{1}{2}mv^2$ or $0.86 = \frac{1}{2}mv^2$ $(m =) 0.72^2 / (2 \times 0.86) = 0.30 \text{ (kg)}$ or $v = 2E_k / p$ $v = (0.86 \times 2) / 0.72 = 2.4 \text{ (to 2 s.f.)}$ $m = 0.72 / 2.4 = 0.30 \text{ (kg)}$
(c)	$(\Delta)E = mg(\Delta)h$ $h = 0.86 / (0.30 \times 9.81)$ $= 0.29 \text{ m}$
(d)	$\cos \theta = (0.93 - 0.29) / 0.93$ so $\theta = 47^\circ$
(e)	$\text{moment} = (0.30 \times 9.81) \times 0.93 \times (\sin 47^\circ \text{ or } \cos 43^\circ)$ or $\text{moment} = (0.30 \times 9.81) \times [0.93^2 - (0.93 - 0.29)^2]^{0.5}$ $= 2.0 \text{ N m}$

32.

(a)	force $\times$ distance
	perpendicular distance of (line of action of) force from the point
(b)(i)	$\text{distance moved by pointer} = 123 - 86 (= 37 \text{ mm})$ $(\text{extension} =) 37 \times (1.8 / 52.6) = 1.3 \text{ (mm)}$ or $\sin \text{ or } \tan \theta = 37 / 526 \text{ (so } \theta = 4.0^\circ \text{ so extension} =) \sin \text{ or } \tan \theta \times 18 = 1.3 \text{ (mm)}$
(b)(ii)	$\text{moment} = 0.472 \times 9.81 \times 6.2 \times 10^{-2}$ $= 0.29 \text{ N m}$
(b)(iii)	$(\Delta)F \times 1.8 \times 10^{-2} = 0.29$ $\Delta F = 16 \text{ N}$
(b)(iv)	$k = F / x$ $= 16 / (1.3 \times 10^{-3})$ $= 1.2 \times 10^4 \text{ N m}^{-1}$

33.

(a)	$m = \rho V$ or ρAL
	$W = mg$
	$(A =) 24 / (9.81 \times 850 \times 0.18) = 0.016 \text{ (m}^2\text{)}$
	or
	$P = F / A$
	$P = \rho gh$
	$(A =) 24 / (9.81 \times 850 \times 0.18) = 0.016 \text{ (m}^2\text{)}$
(b)(i)	(upthrust =) $24 + 8(.0) = 32 \text{ (N)}$
(b)(ii)	$(\Delta)p = 32 / 0.016 (= 2000)$
	$(\Delta)p = \rho g(\Delta)h$
	$\rho = 2000 / (9.81 \times 0.17)$
	$= 1200 \text{ kg m}^{-3}$
(c)(i)	$E = \frac{1}{2}Fx$ or $E = \frac{1}{2}kx^2$ or $E = \text{area under graph}$
	$(\Delta)E = (\frac{1}{2} \times 8.0 \times 0.40) - (\frac{1}{2} \times 4.0 \times 0.20)$
	or
	$(\frac{1}{2} \times 20 \times 0.40^2) - (\frac{1}{2} \times 20 \times 0.20^2)$
	or
	$\frac{1}{2} \times (4.0 + 8.0) \times 0.20$
	$= 1.2 \text{ J}$
(c)(ii)	length = 30 cm

34.

(a)	point where (all) the weight (of the object) is taken to act
(b)(i)	moments about P are: $(F \times 1.10)$, (44.0×0.60) and (3.0×2.90)
	<i>1 mark for any one correct moment and 2 marks for two correct moments</i>
	$(F \times 1.10) = (44.0 \times 0.60) + (3.0 \times 2.90)$
	$F = 32 \text{ N}$
(b)(ii)	force = $32 + 44 + 3$
	$= 79 \text{ N}$
(c)(i)	$(F = \rho gV)$
	$V = 2.5 / (1100 \times 9.81) (= 2.32 \times 10^{-4})$
	$V = (4/3)\pi r^3$
	$r = [(3 \times 2.32 \times 10^{-4}) / 4\pi]^{1/3}$
	$r = 0.038 \text{ m}$

(c)(ii)	resultant moment = 2.5×2.9
	= 7.3 N m
	or
	resultant moment = $F \times 1.1 - (44 \times 0.6 + 0.5 \times 2.9)$
	= 7.3 N m (allow 7.2 N m or 7.4 N m from rounded values of F)
	direction: anticlockwise

35.

(a)	$F = \rho g V$
	$0.071 = 1.2 \times 9.81 \times V$
	$(V = 6.03 \times 10^{-3} \text{ m}^3)$
	$\frac{4}{3} \times \pi \times r^3 = 6.03 \times 10^{-3}$
	$r = 0.11 \text{ m}$
(b)	$m = 0.053 / 9.81$
	$(= 5.4 \times 10^{-3} \text{ kg})$
	$F = 0.071 - 0.053$
	$(= 0.018 \text{ N})$
	$a = (0.071 - 0.053) / (0.053 / 9.81)$
	$= 3.3 \text{ m s}^{-2}$
(c)(i)	$v^2 = u^2 + 2as$
	$3.6^2 = (-1.4)^2 + 2 \times 9.81 \times s$ or $3.6^2 = 1.4^2 + 2 \times 9.81 \times s$
	$s = 0.56 \text{ m}$
(c)(ii)	single straight line from any positive non-zero value of v at $t = 0$ to any negative non-zero value of v at $t = 7$
	line starting at $(0, 1.4)$ and ending at $(7, -3.6)$

36.

(a)	point where (all) the weight (of an object) is taken to act
(b)(i)	vertical component = $45 \sin 37^\circ$ = 27 N
(b)(ii)	the magnitudes of the three moments about A are (23×0.48) , (27×0.56) and $(W \times 0.76)$ correct magnitude of any one moment about A correct magnitudes of any two moments about A $(23 \times 0.48) + (W \times 0.76) = 27 \times 0.56$ $W = 5.4 \text{ N}$
(b)(iii)	horizontal component = $45 \cos 37^\circ$ = 36 N
(b)(iv)	decrease
(b)(v)	$\sigma = F / A$ $\sigma = F / \pi r^2$ or $4F / \pi d^2$ so $\sigma = 5.3 \times 10^7 \times \pi r^2 / \pi (3r)^2$ = $5.3 \times 10^7 / 9$ = $5.9 \times 10^6 \text{ Pa}$

37.

(a)	sum of CW moments = sum of ACW moments about the same point for (an object in rotational) equilibrium
(b)	moment = $0.3(0) \times 0.29 \cos 40^\circ$ or $0.3(0) \times 0.222$ = 0.067 Nm
(c)(i)	$k = F / x$ or $k = \text{gradient}$ e.g. $k = 21 / 10 \times 10^{-3}$ $k = 2100 \text{ N m}^{-1}$
(c)(ii)	$V_{\text{sphere}} = \frac{4}{3} \times \pi \times (0.0480)^3$ $F = \rho g V$ (upthrust =) $1000 \times 9.81 \times (\frac{4}{3} \times \pi \times (0.048)^3) \times 0.26(0) = 1.18 \text{ (N)}$
(c)(iii)	1.18×0.29 or 0.30×0.29 or $F \times 0.017$ $(1.18 \times 0.29) = (0.30 \times 0.29) + (F \times 0.017)$ $F = 15 \text{ N}$

(c)(iv)	$E_{(P)} = \frac{1}{2}kx^2$ or $E_{(P)} = \frac{1}{2}Fx$ $x = F/k = 15/2100$ or x determined from graph for $F = 15.0$ N $E_P = \frac{1}{2} \times 2100 \times (15/2100)^2$ or $E_P = \frac{1}{2} \times 15 \times (15/2100)$ $E_P = 0.054$ J
(d)	the <u>sphere</u> has gained gravitational potential energy

38.

(a)	resultant force (in any direction) is zero resultant moment/torque (about any point) is zero
(b)	(component =) $17\sin 50^\circ = 13$ (N) or $17\cos 40^\circ = 13$ (N)
(c)	$(W \times 0.25)$ or (12×0.35) or (13×0.50) $(W \times 0.25) + (12 \times 0.35) = (13 \times 0.50)$ $W = 9.2$ N
(d)	$F = 9.2 + 12 - 13$ $= 8$ N
(e)	decrease

39.

(a)	$\rho = m/V$ or $\rho = m/Ah$ $p = F/A$ or $p = W/A$ appropriate algebra leading to $p = \rho gh$ e.g. $p = \rho Ahg/A$ or $\rho Vg/A$ or $\rho Vg/(V/h)$ <u>and</u> (so) $p = \rho gh$
(b)	there is atmospheric / air pressure
(c)	$\Delta p = \rho g \Delta h$ e.g. $(9.66 - 9.60) \times 10^4 / 8.0 \times 10^{-2} = \rho \times 9.81$ $\rho = 760 - 770$ kg m ⁻³
(d)	$F = \rho g V$ $= 760 \times 9.81 \times 3.7 \times 10^{-4} \times 4.0 \times 10^{-2} (= 0.11$ N) tension = $0.53 - 0.11$ $= 0.42$ N or $F = (\Delta p) \times A$ $= (9.63 - 9.60) \times 10^4 \times 3.7 \times 10^{-4} (= 0.11$ N) tension = $0.53 - 0.11$ $= 0.42$ N